TEAM BOOGIE

P1: Needfinding

Disability Golf (DisGo) Cart Service

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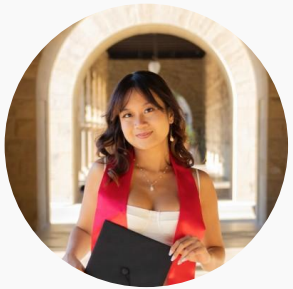
Our *team*

Ari B.



Mai H.

Ananya N.

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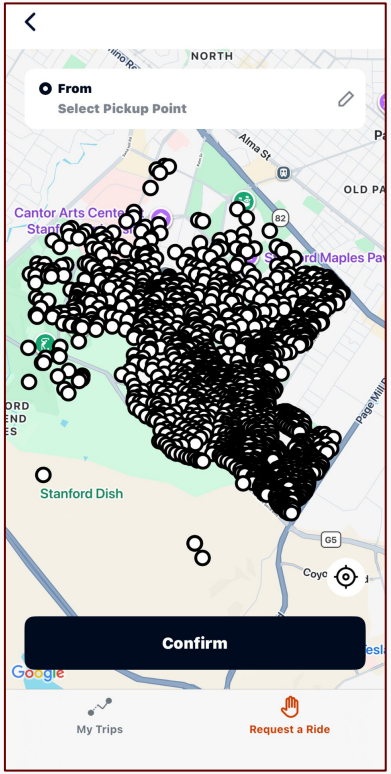
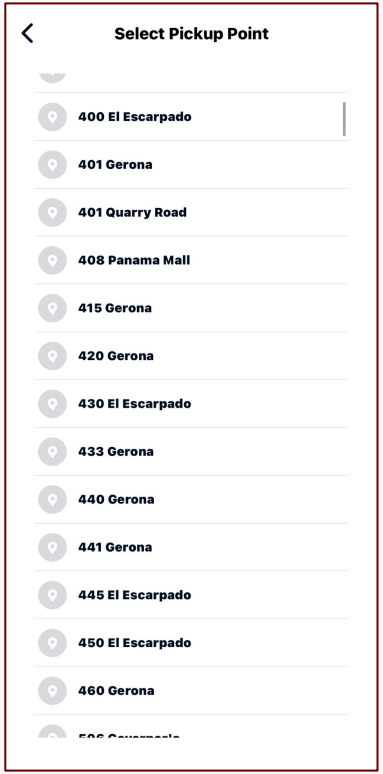
THE PROBLEM SPACE

DisGo (Stanford's disability golf cart service) uses an app (Flexigo) that is NOT currently accessible for BLV users →

incompatible with screen readers





Interviews & Observation

- Interviewed Stanford-based members of the BLV community:
  - **B.S.**: A student who is an active user of DisGo
  - **M.C.**: An instructor (and former Stanford undergrad/coterm student)
- Interviewed and observed DisGo drivers during their working hours of operating the golf carts and conveying students

"Instead of being able to input a search keyword.. [I] have to scroll through a list of 1400 items... So at this point I kind of have this mental model of where my preferred locations are. Around 80% of the scroll bar is where 'O' appears, so 2 above that will get me to Munger, where I live... "

~ Interviewee 1

POV: Stanford student DisGo user

We Met...

A **blind-identifying Stanford coterminal student** who has used DisGo throughout their Stanford career as a mode of accessible transportation.

We Were Surprised To Notice...

How they describe DisGo as "immensely valuable" yet have to **come up with many informal strategies** (memorizing scroll positions, calling drivers, improvising destinations) **to make the system usable**.

We Wonder If This Means...

They are having to **negotiate with themselves** about **what degree of friction is tolerable** within this transportation process in exchange for being able to accomplish their goal.

It Would Be Game Changing To...

Shift the burden of adaptation from the user **to the interface/scheduler** and reduce the need for users to negotiate between effort and independence.

"I'm very much [an] exploratory traveler, I tend to know my surroundings... and go from place to place... [to] just look around and understand the landmarks and the surroundings"

"If it was the first time I was going to NVIDIA... I would probably... before the quarter started... go there ... [and] explore that route."

~ Interviewee 2

POV: Stanford instructor, non-DisGo user

We Met...

A **Stanford instructor** who **identifies as blind or visually impaired**, who prefers **public transportation** to rideshare services, both on and off campus.

We Were Surprised To Notice...

That they prefer a **longer commute** filled with **exploration** and **self-navigation** over a more efficient commute that doesn't leave room for spatial exploration, but rarely ventures to new destinations.

We Wonder If This Means...

They have needed to **contain their exploration to places accessible by public transportation** based on the resources they feel comfortable using.

It Would Be Game Changing To...

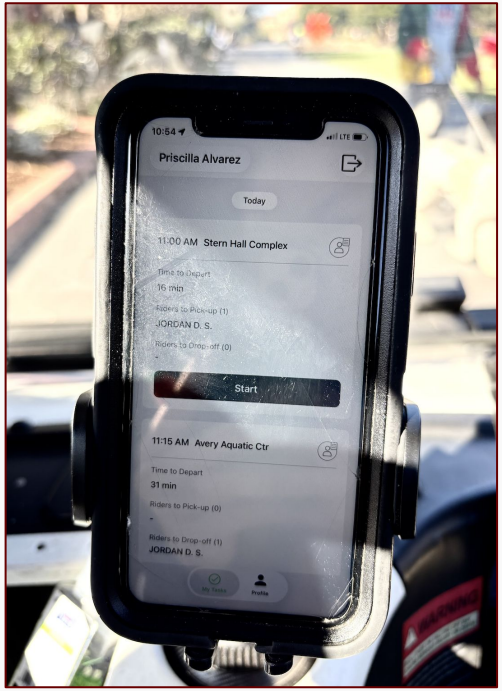
Support mobility that preserves opportunities for **exploratory learning** and spatial understanding and **allows BLV** users **to utilize** existing **mental maps and environmental cues.**

Themes

- Compromise is a constant → resources define degree of independence
- Simplicity and organization are *key accessibility allies*, especially for screen reader users
- Routine/consistency is a good starting point that helps with environmental mapping (but shouldn't be the only use case)
- Cues in other shapes and forms become very valuable

DisGo Driver Interview Insights

- We conducted **three ride-alongs**, observing DisGo drivers and the students they drove.
- We discovered that…
  - Drivers often meet students at places different from where the app indicates
  - Not all the buildings are on the searchable list
    - Lacks specificity (i.e. specific dorms, front door/back door, etc.)
  - Drivers cannot see rider's PIN



Next Steps

- Brainstorm solutions based on our findings
- Build an initial prototype following WCAG AA standards
- Find more BLV users to interview and test our prototype

Thank you.

Questions?